JMeter Performance Test Report: ParaBank

Group 40 Performance Testing with JMeter

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1. Purpose

This document evaluates the performance of ParaBank demo banking portal (https://parabank.parasoft.com) while more users access it simultaneously. Apache JMeter 5.6.3 was employed to simulate one user journey for different levels of load consisting of 1, 10, 20, and 50 virtual users. The journey was:

Login → Accounts Overview → View Account Details → Transfer Funds → View Transactions → Logout

2. How the test was set up

All four loads are in one test plan file (jmeterplan-40.jmx). The main settings are below.

Setting	Value and reason
Thread Groups	Four groups: 1, 10, 20 and 50 Users
Ramp-up period	10 seconds for every group, so users start gradually.
Duration	20 seconds per group (Specify Thread lifetime, loop count Infinite)
Independent runs	"Run Thread Groups consecutively" is ticked, so each group runs on its own, one after another
On sampler error	Continue, so one failed request does not stop the test
300 ms delay	Constant Timer (300 ms) placed under the Home Page request, so it runs before the page loads
Think time	Uniform Random Timer on the whole plan: 500 ms fixed plus up to 500 ms random between steps, to act like a real user pausing between clicks
Session handling	HTTP Cookie Manager keeps the login cookie (JSESSIONID); cookies clear at the start of each loop
Server address	HTTP Request Defaults: https, parabank.parasoft.com
Test data	User Defined Variables for username, password, fromAccount and toAccount, so a database reset only needs four values changed
Correlation	A Regular Expression Extractor reads the customer ID from the Accounts Overview page for the next request
Listeners	View Results Tree and Aggregate Report under each Thread Group

2.1 Requests and assertions

Every request has at least one assertion. We used several assertion types so each check suits the response it is testing.

Request	Method and path	Assertion(s)
01 Home Page	GET /parabank/index.htm	Response Assertion: text contains "Customer Login"

Request	Method and path	Assertion(s)
02 Login	POST /parabank/login.htm	Response Assertion: text contains "Accounts Overview"
03 Accounts Overview	GET /parabank/overview.htm	Duration Assertion (under 10,000 ms); Response Assertion
03b Load accounts list	GET .../customers/{id}/accounts	JSON Assertion: $[0].id exists
04 View Account Details	GET /parabank/activity.htm?id=...	Response Assertion; Size Assertion (over 1000 bytes)
05a Transfer Funds page	GET /parabank/transfer.htm	Response Code Assertion (equals 200); Response Assertion
05b Submit transfer ($1)	POST .../bank/transfer?...	Response Assertion: "Successfully transferred"
06 View Transactions	GET .../accounts/{id}/transactions	JSON Assertion: $[0].amount exists
07 Logout	GET /parabank/logout.htm	Response Assertion with NOT ("Log Out" must be gone); Response Assertion

03b, 05b and 06 are the AJAX calls that the browser will make on the pages. This inclusion implies that the test replicates the requests made by an actual browser.

3. Results

The table below shows the TOTAL row from each Aggregate Report. Times are in milliseconds.

Users	Samples	Average	Median	Min	Max	95% line	Error %	Throughput
1	15	525	505	288	834	788	0.00%	47.8/min (0.8/s)
10	102	688	421	62	3,996	2,062	18.63%	5.4/s
20	320	152	102	54	842	403	100.00%	16.8/s
50	459	965	505	62	13,588	3,714	100.00%	18.1/s

The two shaded rows had every request rejected. Section 4 explains why their response times should not be read the same way as the 1 and 10 user results.

3.1 Detail for 10 users

The 10-user run is the most useful one, because it shows the change from working to failing. The busiest requests were the slowest:

Request	Average	95% line	Max	Error %
01 Home Page	1,064	2,821	3,514	18.75%
02 Login	715	994	2,766	21.43%
04 View Account Details	568	1,115	1,683	9.09%
06 View Transactions	438	856	856	33.33%
07 Logout	1,355	3,996	3,996	37.50%

Error % rises for the steps later in the journey. Users who started first finished cleanly. Users who reached Transactions and Logout near the end of the 20 seconds were more likely to be blocked.

4. Analysis

4.1 Throughput

At one user level, the throughput rate was 47.8 transactions per minute (approximately 0.8 transactions per second). This is deliberately kept low since the single user spends 300 milliseconds waiting for the home page and 0.5 to 1 seconds waiting between each step. At ten users, it increased to 5.4 transactions per second, which was expected from the number of users.

At twenty and fifty users, throughput seems even higher (16.8 and 18.1 transactions per second), but this is deceptive because all of those transactions failed. The server answered quickly with a short error, so users moved through the loop faster and sent more requests. More throughput here does not mean more useful work was done. The jump from 20 to 50 users only added about 8% throughput, which shows the system had already hit a ceiling.

4.2 Average and median response time

The mean went up from 525 ms (1 user) to 688 ms (10 users). The median was lower (from 505 to 421 ms) although the mean went up. This indicates that many transactions were quick although there were some extremely slow transactions. This is a typical early sign of a server starting to struggle.

At 20 users the average dropped to 152 ms. This is not an improvement. The responses were 429 errors with a 17-byte body, which take almost no work to send. At 50 users the average rose again to 965 ms, even though everything was still being rejected. So the requests were waiting in a queue before they were even refused.

4.3 Minimum and maximum response time

The minimum dropped from 288 ms with 1 user to 54 – 62 ms in all other tests. These really low figures coincide with the refused requests since the rejection process takes less time than the creation of a real web page.

The maximum increased greatly: 834 ms (1 user), 3,996 ms (10 users) and 13,588 ms (50 users). The 13.6-second maximum was on the Home Page, the first step every user hits. With 50 users all starting within 10 seconds, that page took the biggest burst of traffic.

4.4 95th percentile

The 95th percentile shows what the slowest 5% of users experienced. It went from 788 ms (1 user) to 2,062 ms (10 users) and 3,714 ms (50 users). For the Home Page at 50 users it was 8,704 ms. So one user in twenty waited almost 9 seconds just for the first page, which most people would find unacceptable.

4.5 Error %

This is the clearest change in the whole test:

- **1 user:** 0% errors. Every step passed every assertion.
- **10 users:** 18.63% errors. The first loops passed, then failures started part way through.
- **20 and 50 users:** 100% errors. Every request was rejected.

HTTP 429 errors (Figures 7 and 8) occurred. The HTTP headers reveal that the website is protected by Cloudflare and have the "Retry-After" field with an estimate of 280-300 seconds. That is to say, after the website detects that the client is sending too many requests, it bans the client from accessing the website for five minutes.

5. Why these results happened

- **Rate limiting, not a crash.** ParaBank (through Cloudflare) protects itself by refusing extra traffic instead of slowing down for everyone. That is why the error rate jumped so sharply instead of climbing slowly.
- **All virtual users come from one computer.** The rate limit counts requests per client address. From the server's side, 50 users on one laptop look like a single very busy visitor.
- **The block carried over between groups.** The groups ran back to back. The 10-user group triggered the limit, and the five-minute Retry-After meant the 20-user group started while the block was still active. This is why the 20-user group failed from its very first request.
- **One shared test account.** All virtual users logged into the same account and moved money between the same two accounts. The brief allows this, but on a real system it could cause locking or contention on that account.
- **Shared public server.** ParaBank is a free demo used by many students and testers at the same time, so its speed also depends on traffic we cannot see or control.
- **Network distance.** Tests ran from Melbourne to a remote server. Part of every response time is just travel time over the internet.

6. Problems found during testing

- **Database reset.** Part way through, ParaBank wiped its data and our test account disappeared. Login started failing with "The username and password could not be verified." We registered again and only had to update the four variables. This is exactly why we used variables instead of typing values into each request.
- **429 blocking while building the test.** Early runs with no think time sent requests too fast and were blocked. We added a think-time timer (0.5 to 1 second between steps). This made the test more realistic and let the full journey pass with 1 user.

7. Conclusion

ParaBank handled one user well: every step passed, with an average of 525 ms and a 95th percentile under 0.8 seconds. With 10 users it was still mostly working, but slower and less predictable, and close to one in five requests failed. At 20 and 50 users every request was rejected with 429 Too Many Requests.

So the main limit we found is not page speed but the site's rate limit, which sits somewhere between 10 and 20 users from one source in our setup. For a fairer measurement of the server itself, future tests could run each load separately with a break of at least five minutes in between, spread users across more than one machine or address, and use more than one test account.

Appendix: Screenshots

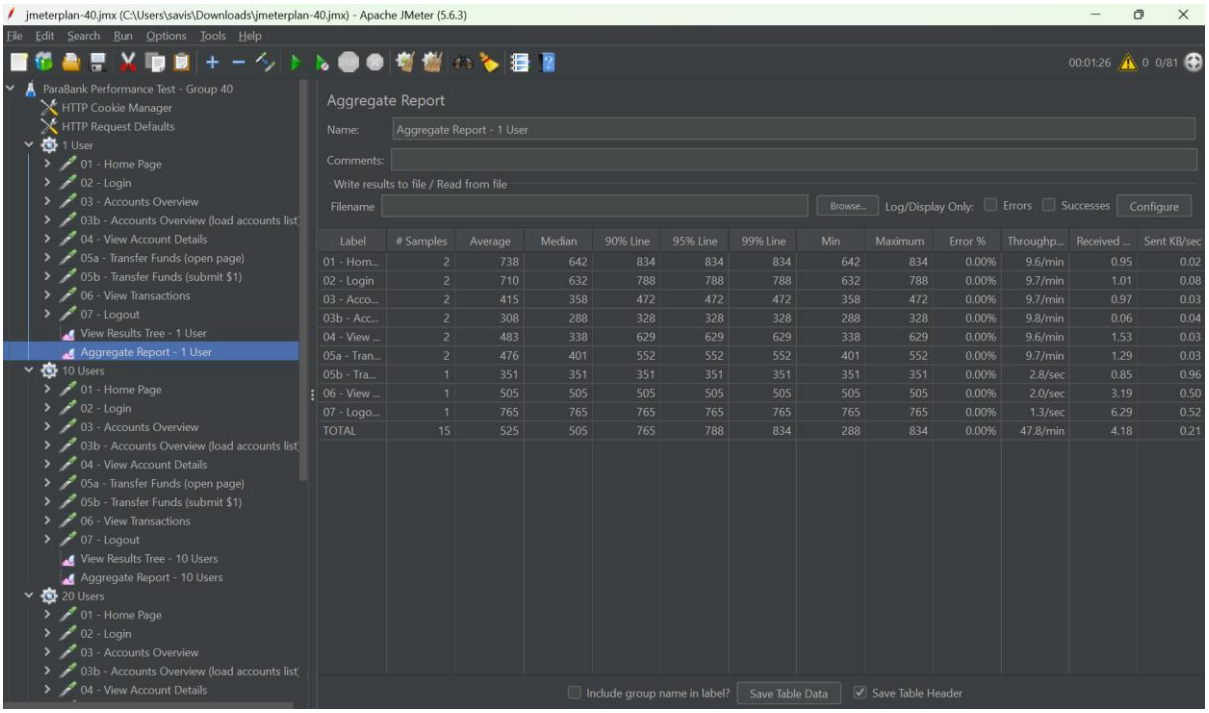


Figure 1: Aggregate Report, 1 user

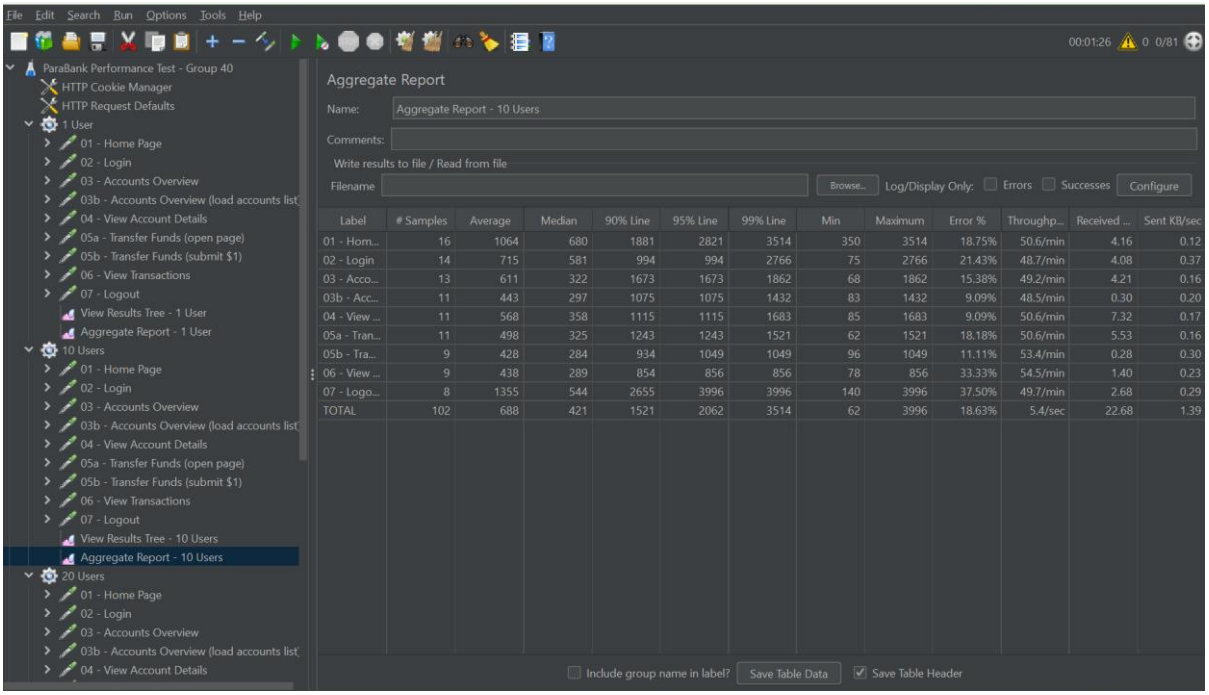


Figure 2: Aggregate Report, 10 users

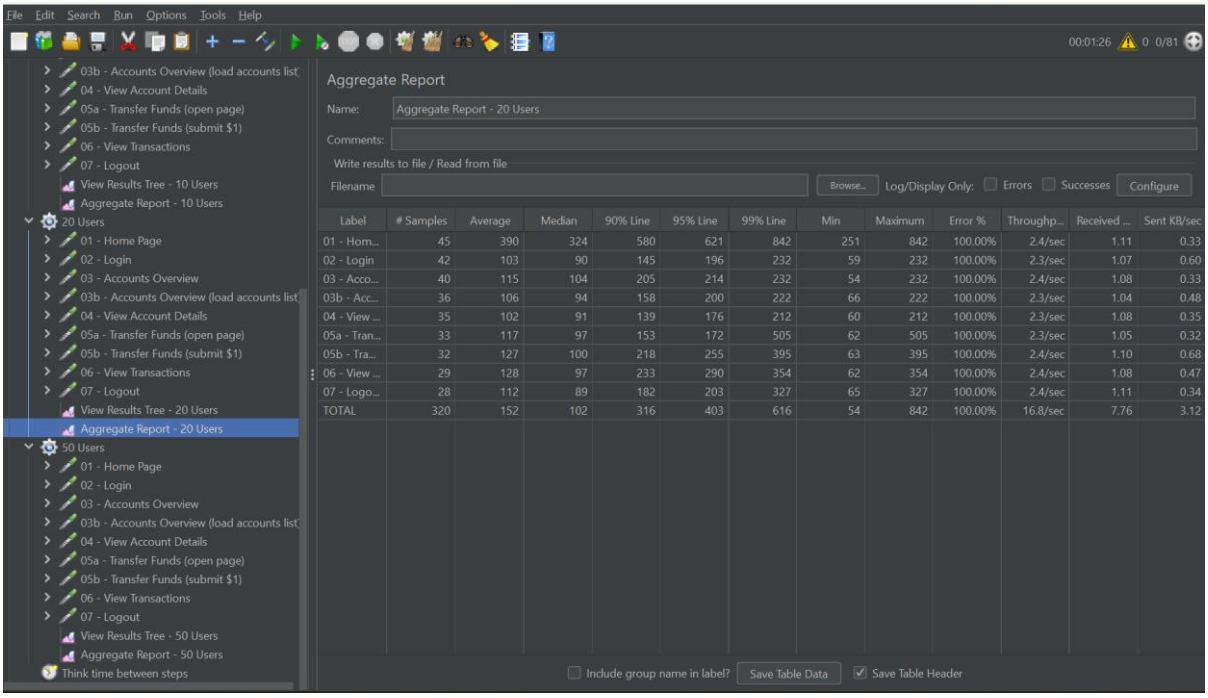


Figure 3: Aggregate Report, 20 users

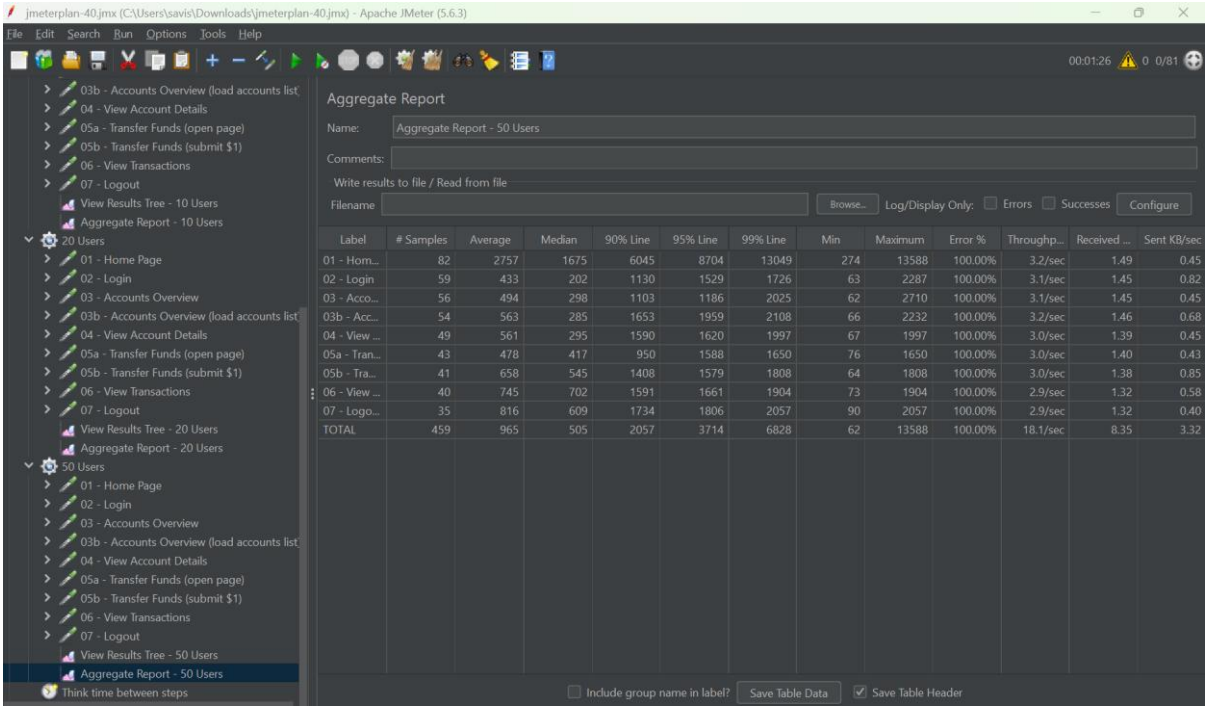


Figure 4: Aggregate Report, 50 users

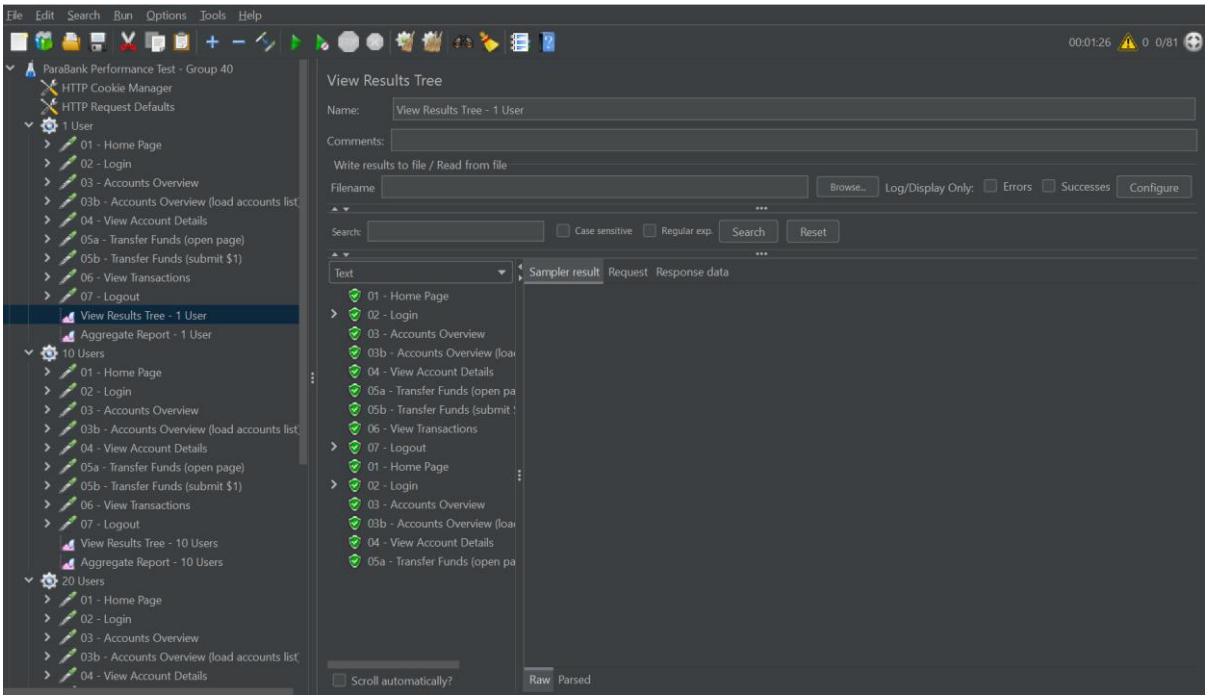


Figure 5: View Results Tree, 1 user: every step passes

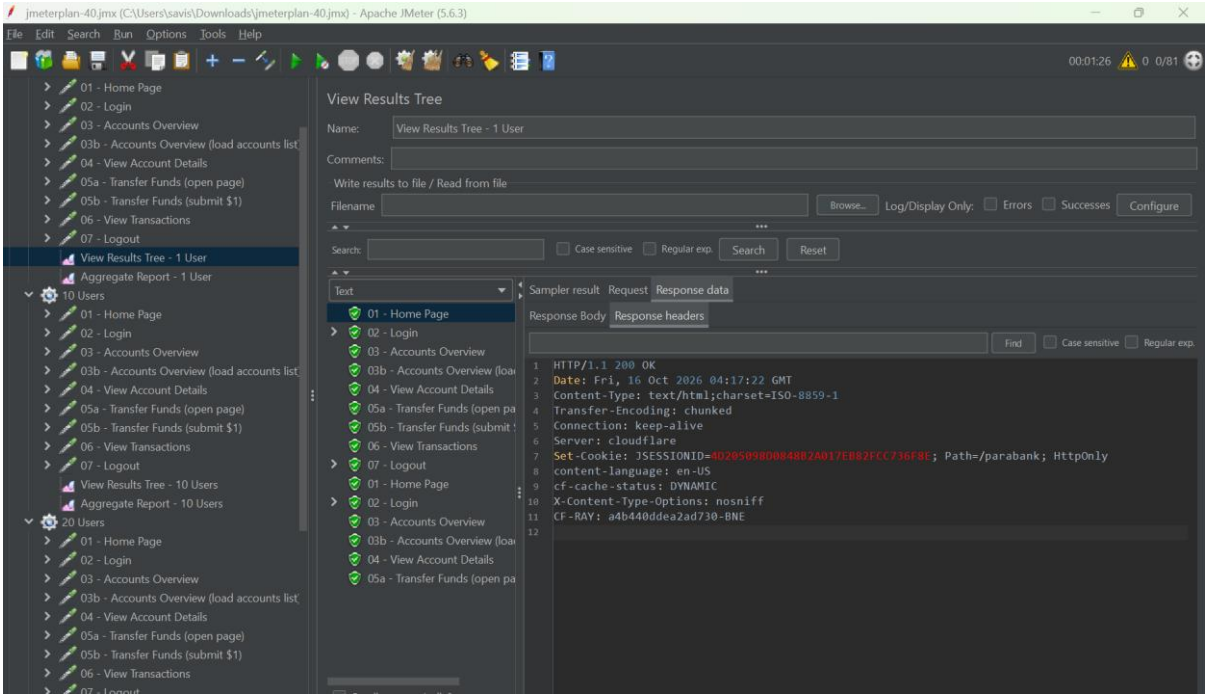


Figure 6: Response headers for a successful request (200 OK)

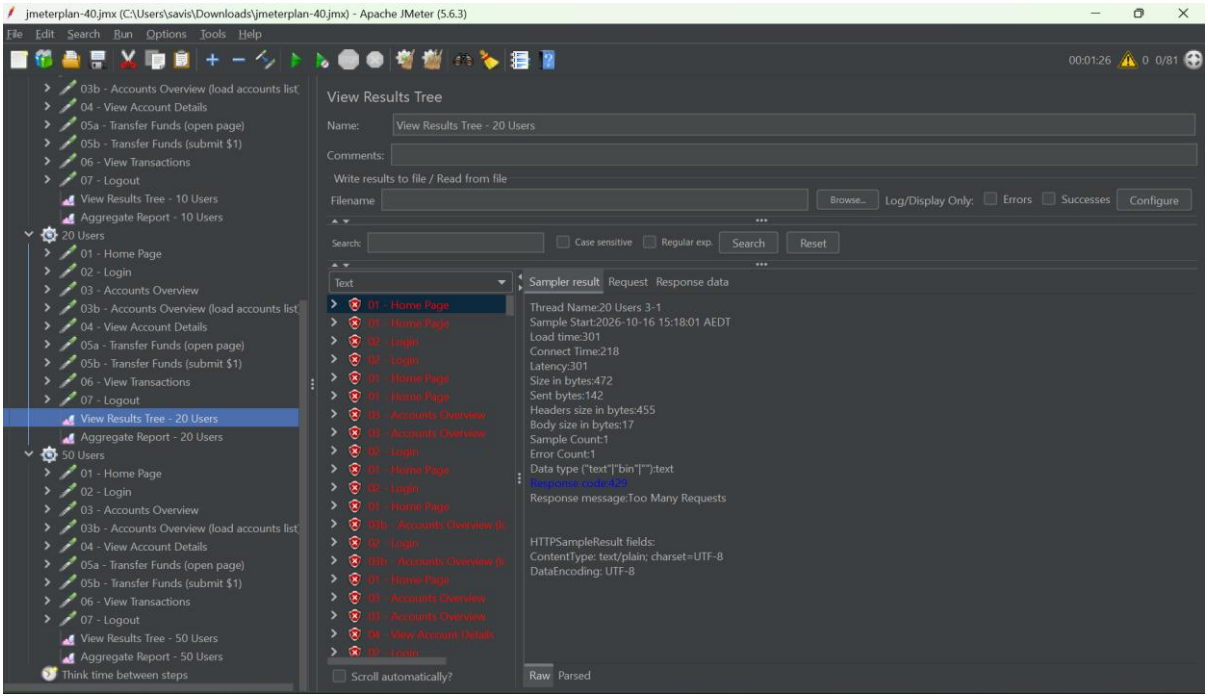


Figure 7: A rejected request: response code 429 Too Many Requests

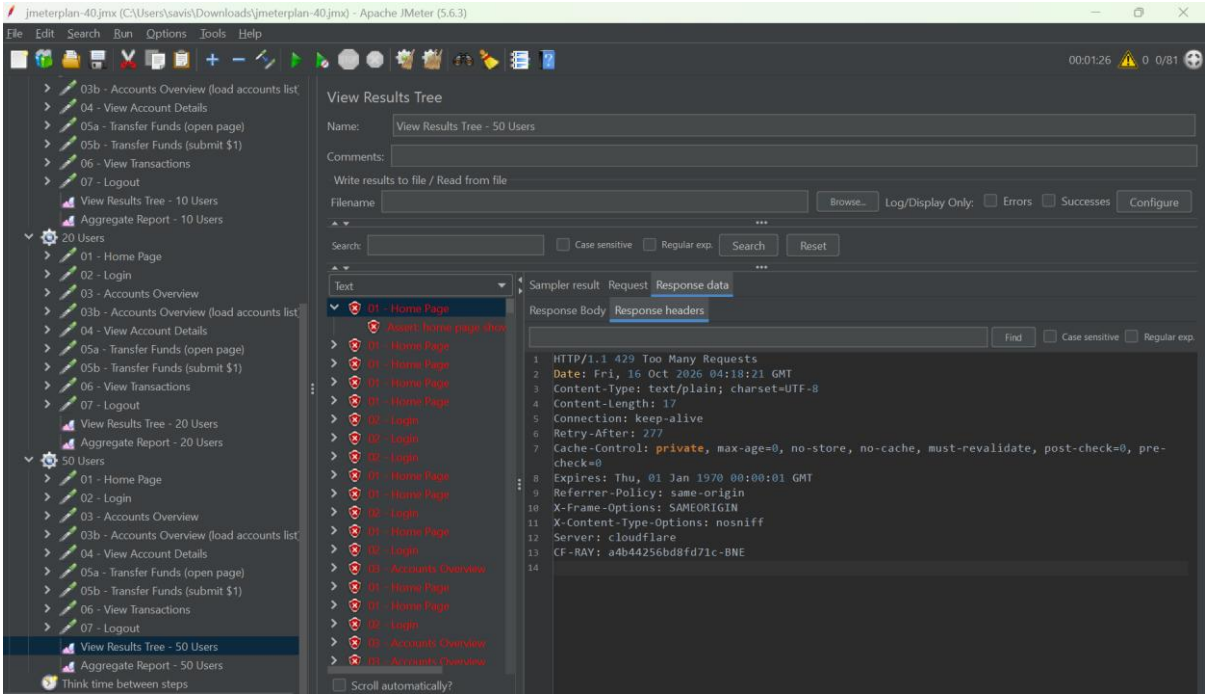


Figure 8: Response headers for a 429 response, showing Retry-After and Server: cloudflare

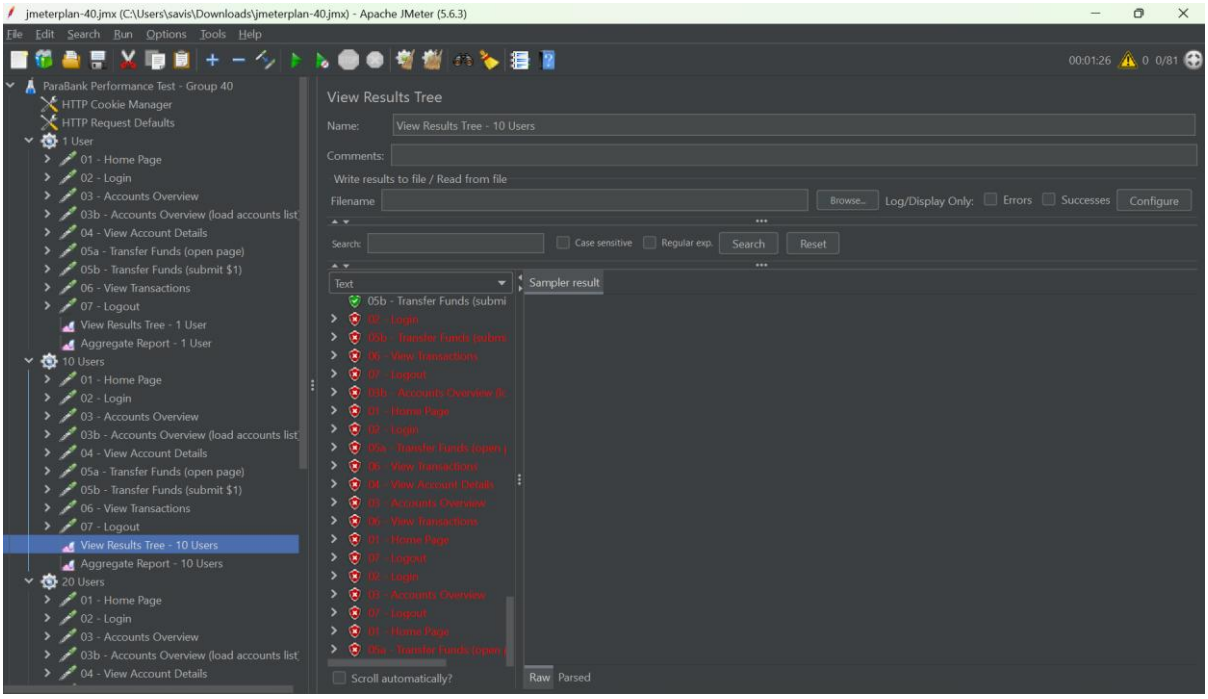


Figure 9: View Results Tree, 10 users: requests start passing, then begin to fail

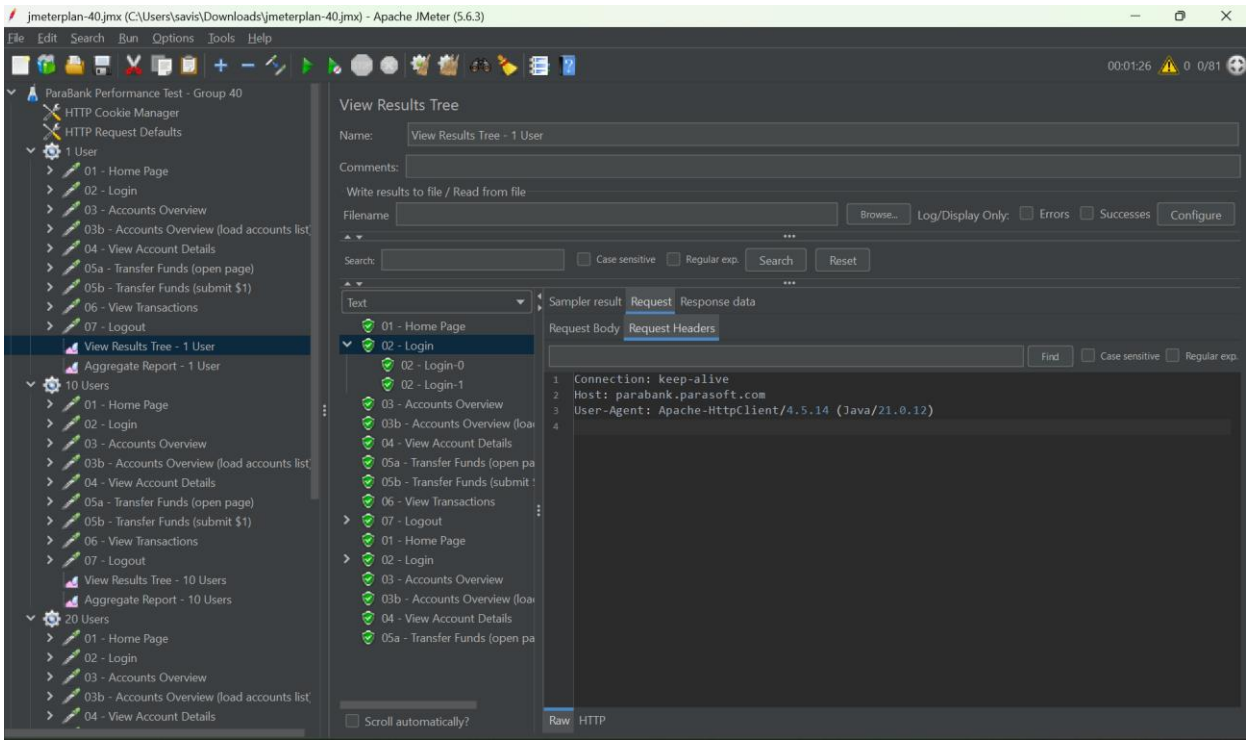


Figure 10: 02 Login Request Tab